

Installing R and RStudio

STAT 1210-2 — Introduction to Statistics for the Life and Social Sciences

University of Prince Edward Island
School of Mathematics and Computational Sciences

Fall 2026

Before You Begin

We will occasionally use the programming language **R** in STAT 1210 to perform statistical calculations and work with data. No previous programming experience is required.

For this course, you should install two programs:

1. **R** — the programming language that performs the statistical calculations.
2. **RStudio** — a program that provides a more convenient environment for writing and running R code.

Important

You must install **R first** and then install **RStudio**. RStudio requires R to be installed on your computer.

Both R and RStudio Desktop are available free of charge.

Step 1: Install R

R should be downloaded from the Comprehensive R Archive Network (CRAN):

<https://cran.r-project.org/>

Choose the instructions below that correspond to your operating system.

Windows

1. Open the CRAN website: <https://cran.r-project.org/>
2. Select **Download R for Windows**.
3. Select **base**.
4. Select the link to download the latest version of R for Windows.
5. Once the installer has downloaded, open the file.
6. Follow the installation instructions. The default installation options are appropriate for this course.
7. When the installation is complete, close the installer.

You do not need to change the default installation folder or customize the installation.

macOS

1. Open the CRAN website: <https://cran.r-project.org/>
2. Select **Download R for macOS**.
3. Download the current R installer appropriate for your Mac.

4. Open the downloaded `.pkg` file.
5. Follow the installation instructions.
6. The default installation options are appropriate for this course.

If the download page provides different installers for Apple Silicon and Intel Macs, choose the version appropriate for your computer.

Step 2: Install RStudio

After R has been installed, install **RStudio Desktop**.

The official RStudio download page is:

<https://posit.co/download/rstudio-desktop/>

RStudio is developed by **Posit**. You may therefore see the names *Posit* and *RStudio* on the same website.

Windows

1. Open the RStudio Desktop download page.
2. Download the free version of **RStudio Desktop** for Windows.
3. Open the downloaded installer.
4. Follow the installation instructions.
5. The default installation options are sufficient for this course.
6. After installation is complete, open **RStudio**.

macOS

1. Open the RStudio Desktop download page.
2. Download the free version of **RStudio Desktop** for macOS.
3. Open the downloaded `.dmg` file.
4. Drag the RStudio application into the **Applications** folder when prompted.
5. Open RStudio from the Applications folder.

Step 3: Check That Everything Works

Open **RStudio**.

You should see several panels. One of these panels is called the **Console**. The Console is where R commands can be entered and run.

Find the > symbol in the Console and enter:

```
2 + 2
```

Press **Enter**.

R should return:

```
[1] 4
```

If you see this result, R and RStudio are working correctly.

Next, try:

```
sqr(25)
```

The result should be:

```
[1] 5
```

Finally, try creating a small collection of data and calculating its mean:

```
x <- c(2, 4, 6, 8, 10)
mean(x)
```

R should return:

```
[1] 6
```

You are now ready to use R for STAT 1210.

Creating an R Script

Although commands can be entered directly into the Console, most of the R code used in this course will be written in an **R script**.

To create a new script in RStudio:

1. Select **File** from the menu at the top of RStudio.
2. Select **New File**.
3. Select **R Script**.

A new editor window will appear. Enter:

```
x <- c(2, 4, 6, 8, 10)

mean(x)
median(x)
sd(x)
```

To run a line of code:

1. Place the cursor on the line that you want to run.

2. Select **Run** near the top of the script editor.

You can also use the keyboard shortcut:

- **Windows:** Ctrl + Enter
- **macOS:** Command + Enter

The result will appear in the Console.

Saving Your R Script

R scripts use the file extension `.R`. For example:

```
stat1210-practice.R
```

You should create a folder on your computer for STAT 1210 and save your R scripts in that folder. For example:

```
STAT1210
```

You may create subfolders later for lectures, assignments, and other course material.

R Packages

R can be extended using collections of additional functions called **packages**. We may (highly unlikely) use several packages during the course. If a package is required, installation instructions will be provided in class. In general, a package is installed using:

```
install.packages("package_name")
```

For example:

```
install.packages("ggplot2")
```

A package normally needs to be installed only **once** on your computer.

After installation, the package can be loaded into the current R session using:

```
library(ggplot2)
```

Unlike installation, the `library()` command normally needs to be run again when you start a new R session and want to use that package.

Common Installation Problems

RStudio Opens but R Does Not Work

Make sure that **R was installed before RStudio**. If necessary, install R from CRAN and then restart RStudio.

Your Computer Prevents the Installer from Opening

Make sure that you downloaded R from the official CRAN website and RStudio from the official Posit website. Your operating system may ask you to confirm that you want to open an application downloaded from the Internet.

You Have an Older Version of R

For this course, using a reasonably current version of R is recommended. If R is already installed on your computer and RStudio works correctly you may not need to reinstall it immediately.

You Are Still Having Problems

If R or RStudio does not install correctly, contact me rightaway or bring your device to me during office hours. Include a screenshot or the complete error message when asking for assistance. This makes installation problems much easier to diagnose.